

11.28 DIELECTRICS

Dielectrics are basically insulators which ordinarily do not contain any free charge carriers for conduction. They, however, contain positive and negative charges which are bounded together and hence could be effected by the electric field. Dielectric materials are classified into two categories; namely polar and nonpolar dielectrics.

11.28.1 Polar Dielectrics

If the centres of gravity of the positive and negative charges in a molecule are separated by a short distance ($\sim 10^{-10}$ m), then the molecule is called polar molecule. As a result, it behaves as a dipole with its dipole moment directed along the line joining positive and negative charge. The dielectrics containing such molecules is known as polar dielectrics. In the absence of external electric field, the net dipole moment of polar dielectrics is zero because of random orientation of various dipoles inside the material.

11.28.2 Non-polar Dielectrics

In case, the centres of gravity of the positive and negative charges in a molecule coincide, the molecule is called non-polar molecule. This results into zero dipole moment of the molecule. The dielectrics containing such non-polar molecules is known as non-polar dielectrics. The net dipole moment of the non-polar material will also be zero in the absence of external electric field.

11.29 DIELECTRIC POLARIZATION

Consider a dielectric bar containing polar or nonpolar molecules. Let us place the bar in an external electric field, say between the plates of a charged capacitor. The presence of the external electric field can cause the shifting of the centres of gravity of the positive and negative charges to convert the non-polar molecules into polar molecules. The result will be the alignment of dipoles along the electric field resulting in a net dipole moment of the dielectric bar Fig. 11.25. The *presence of* positive and negative charges, in the neighbourhood of each other, will cancel the effect of each other within the body of the dielectric. But on the surfaces of the bar at right angles to the external field $\vec{E}_0$, there will appear a net positive and negative charge as shown in Fig. 11.25. The appearance of the net positive and negative charges on the surfaces of the bar will set up an electric field $\vec{E}_p$ within the dielectric in a direction opposite to the applied field. And net electric field within the dielectric will be the vector sum of $\vec{E}_0$ and $\vec{E}_p$, i.e., $\vec{E} = \vec{E}_0 + \vec{E}_p$. As a result, the resultant electric intensity inside the dielectric is diminished, i.e., $E < E_0$ therefore, the actual field acting on the molecules or atoms of a dielectric is E instead of E_0 .

The phenomenon of appearance of the charges on the surfaces of the dielectric bar induced by an external electric field is called polarization. Since dipoles align along the electric field, thus, the process of dipole alignment in a dielectric is also known as polarization. Quantitatively, the polarization P is measured as net electric dipole moment per unit volume, i.e., $P = \frac{\Sigma p}{V}$, where p is the average dipole moment induced on each molecule of a dielectric having volume V . The charge on the plates of the capacitor is termed as free charge and the charge induced on the surfaces of the dielectric is called bound charge.

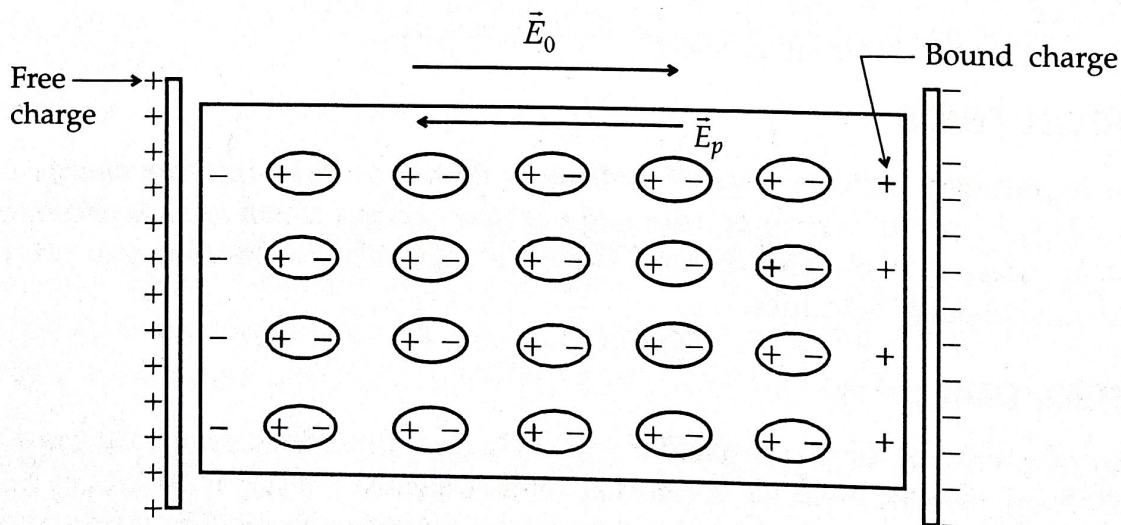


Fig. 11.25 Alignment of Atomic Dipoles when Dielectric Slab is Placed in an Electric Field $\vec{E}_0$

Let l be the length of the dielectric slab and A be the area of cross-section perpendicular to the direction of the field. Due to polarization, let the surface density of charges appearing on the faces be σ_p . The

electric dipole moment will be $\sigma_p Al$, where Al represent the volume of the dielectric bar. If P be the *dielectric polarization*, then electric dipole moment is PAI . Therefore, we have

$$PAI = \sigma_p \cdot AI$$

$$\text{or } P = \sigma_p \quad \dots(43)$$

Hence the surface density of charges appearing on the faces perpendicular to the field is the measure of polarization.

Now the average dipole moment p induced in each molecules of a dielectric is proportional to E , i.e.,

$$p \propto E$$

$$\Rightarrow p = \alpha E \quad \dots(44)$$

where α is a constant and is known as *polarizability of the dielectric*. If there are n dipoles per unit volume in a dielectric then the electric dipole moment per unit volume is $n\alpha E$ and is represented by P , known as *polarization*

$$\therefore P = n\alpha E \quad \dots(45)$$

11.29.1 Types of Polarization

In dielectric materials, polarization occurs due to several atomic mechanisms which results into following four types of polarization:

1. Electronic polarization (P_e)
2. Ionic polarization (P_i)
3. Orientational polarization (P_o)
4. Interfacial or space charge polarization (P_s)

The electronic polarization (P_e) results from a displacement of the centre of the negatively charged electron cloud relative to the positive nucleus of an atom by an electric field. Monoatomic gases exhibit only this type of polarization.

Ionic polarization (P_i) occurs in ionic materials, when an electric field is applied to an ionic material. Cations and anions get displaced in opposite directions and result into a net electric dipole moment. Both these polarizations are temperature independent.

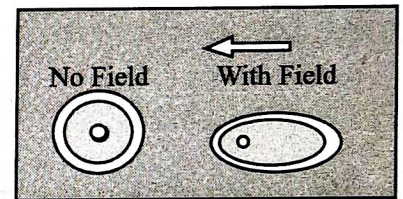


Fig. 11.26 Electronic Polarization

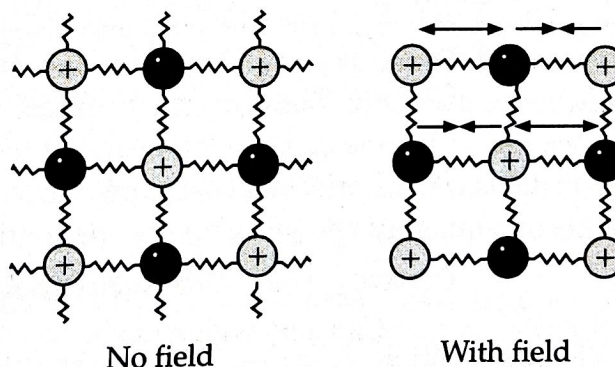


Fig. 11.27 Ionic Polarization

Orientation polarization (P_o), occurs only in those dielectric materials which possess permanent electric dipole moments and can be aligned by the electric field in the same way as magnetic dipoles are aligned by a magnetic field. This polarization is dependent on temperature and decreases with increasing temperature. At low temperature dipole orientation by electric field is opposed by frictional forces.

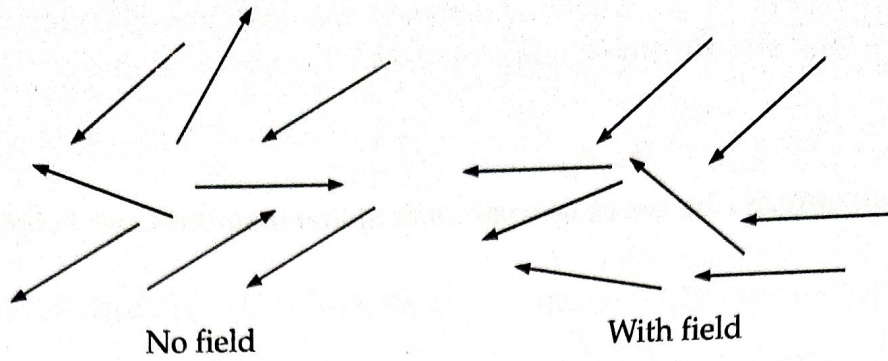


Fig. 11.28 Orientation Polarization

The interfacial or space charge polarization (P_s), occurs due to the accumulation of charges at the interphases in a multiphase dielectric material. Such a polarization is possible when one of the phases present has much higher resistivity than the other. It is found in ferrites and semiconductors and in composite insulators at elevated temperatures.

Therefore, the total polarization, P , of a dielectric substance is equal to the sum of different types of polarization, i.e.,

$$P = P_e + P_i + P_o + P_s.$$

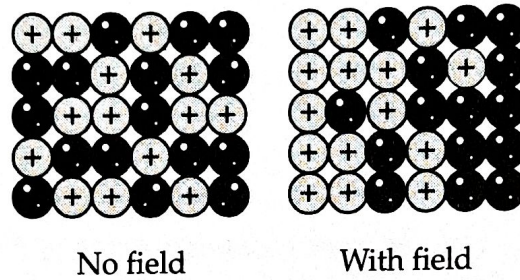


Fig. 11.29 Space Charge Polarization

11.30 DIELECTRIC CONSTANT AND ELECTRIC SUSCEPTIBILITY

It has been observed that in a large number of dielectrics, the polarization P is proportional to the electric field E , i.e.,

$$\begin{aligned} P &\propto E \\ \Rightarrow P &= \epsilon_0 \chi_e E \end{aligned} \quad \dots(46)$$

where χ_e is dimensionless constant called electric susceptibility of the dielectric that measures the amount of polarization a given field produces in a dielectric. The permittivity of free space ϵ_0 , has been introduced to make χ_e dimensionless. Also, we know that the capacity of a parallel plate capacitor increases if the gap between the plates is filled with dielectric material of absolute permittivity ϵ . The increase in capacity of a capacitor is measured in terms of a quantity known as relative permittivity ϵ_r , i.e.,

$$\begin{aligned} \epsilon_r &= \frac{\text{Capacity with dielectric material}}{\text{Capacity with vacuum}} \\ &= \frac{\frac{\epsilon A}{d}}{\frac{\epsilon_0 A}{d}} = \frac{\epsilon}{\epsilon_0} \end{aligned} \quad \dots(47)$$

where A is the area of the plate and d is the spacing between plates of a capacitor. The relative permittivity ϵ_r is also known as dielectric constant of the medium and measures the charge storing capacity of the dielectric.